Project Documentation

Project Title: Superstore Sales Performance Analysis and Customer Insights using Pandas, Matplotlib, Seaborn, Plotly

1. Project Overview

This project focuses on performing Exploratory Data Analysis (EDA) on a Superstore sales dataset to uncover meaningful business insights related to sales performance, profitability, customer segments, product categories, discounts, and regional trends. The analysis helps identify patterns, correlations, and opportunities for improving business decisions through data-driven insights.

The project includes data cleaning, statistical analysis, visualization, and insight generation using Python libraries such as Pandas, NumPy, Matplotlib, Seaborn, and Plotly.

2. Tools Used

- **Python –** Programming language used for analysis
- **Pandas –** Data loading, cleaning, manipulation, and aggregation
- **Matplotlib –** Data visualization and plotting
- **Seaborn –** Statistical data visualization
- **Plotly –** Interactive visualizations
- **Google Colab –** Development environment for analysis and reporting

3. Dataset

- **Source:** Dataset downloaded from public.tableau
- **Description:** The dataset contains structured sales information with the following key columns:
    - Order ID
    - Order Date
    - Ship Date
    - Ship Mode
    - Customer ID
    - Customer Name
    - Segment
    - Country/Region
    - City
    - State/Province

- Postal Code
- Region
- Product ID
- Category
- Sub-Category
- Product Name
- Sales
- Quantity
- Discount
- Profit

4. Steps Followed

1. Data Collection

Loaded the Superstore sales dataset into Google Colab using Pandas.

2. Data Understanding

Explored dataset structure using:

- df.head()
- df.tail()
- df.info()
- df.describe()

3. Data Cleaning

Checked for:

- Missing values
- Duplicate records
- Incorrect data types

4. Feature Engineering

Created derived columns such as:

- Total_Sales
- Total_Profit
- Total_Discount
- Unit_Price
- Unit_Cost
- Profit_Margin
- Region_Segment
- Discount_Rate
- Discount_Amount
- Created Month and Year column in to Order Date column

5. Exploratory Data Analysis (EDA)

Performed univariate, bivariate, and multivariate analysis to identify patterns, relationships, and trends in the dataset. Conducted correlation and trend analysis using GroupBy operations, pivot tables, and statistical summaries to extract meaningful business insights.

6. Data Visualization

Created visualizations to identify:

- Sales trends
- Profit distribution
- Regional performance
- Customer segmentation
- Discount impact

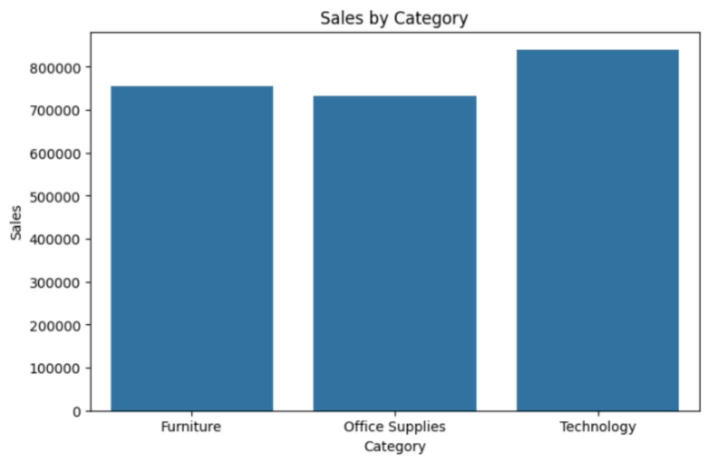
5. Key Insights

- Technology category generated the highest sales and profit.
- The West region showed the strongest overall sales performance.
- High discounts negatively affected profitability.
- Consumer segment contributed the highest number of sales.
- Some products generated losses despite high sales volume.
- Standard Class was the most frequently used shipping mode.
- Sales showed seasonal trends across different months.
- Profit margin decreased significantly when discount rates increased.

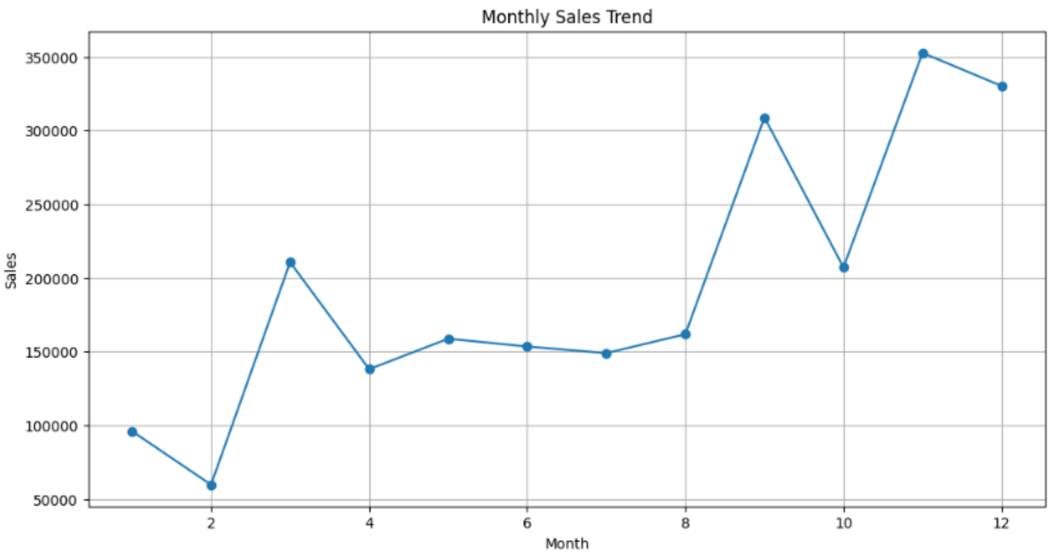
6. Visualizations

The following visualizations were used in the project:

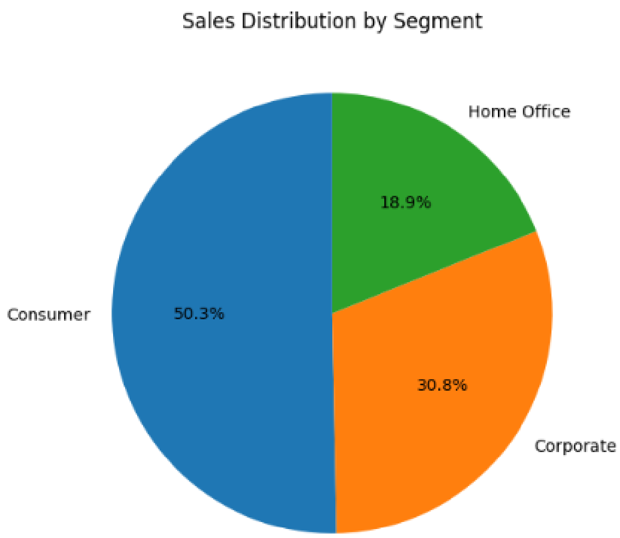
- **Bar Plot –** Compare sales across categories



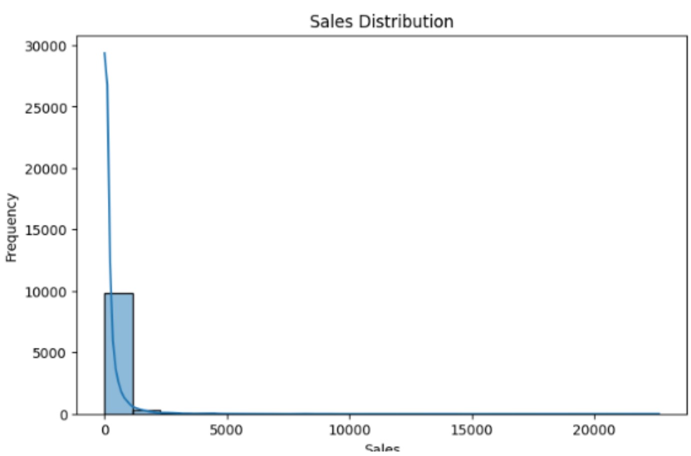
- **Line Chart –** Analyze monthly sales trends



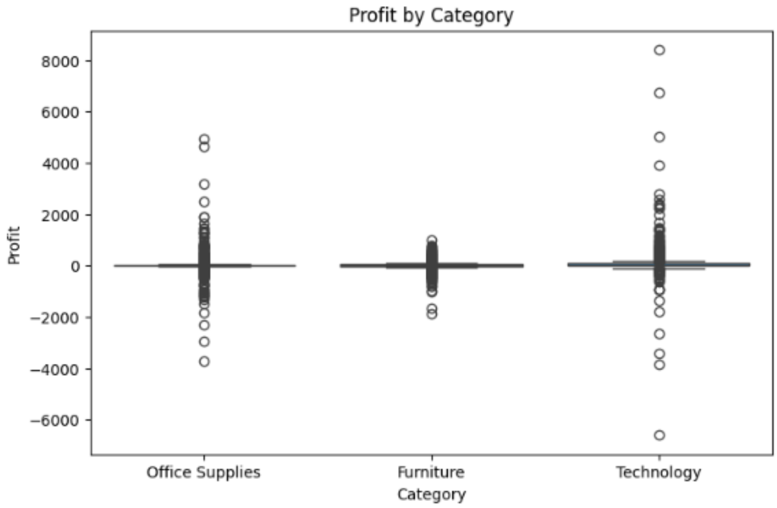
- **Pie Chart –** Show segment-wise sales contribution



- **Histogram –** Understand sales distribution

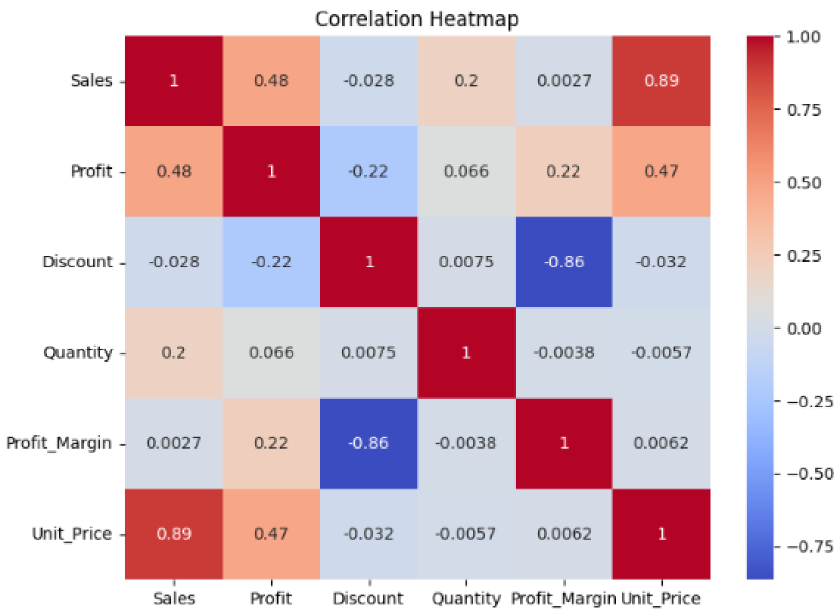


- **Box Plot –** Detect outliers in profit and sales

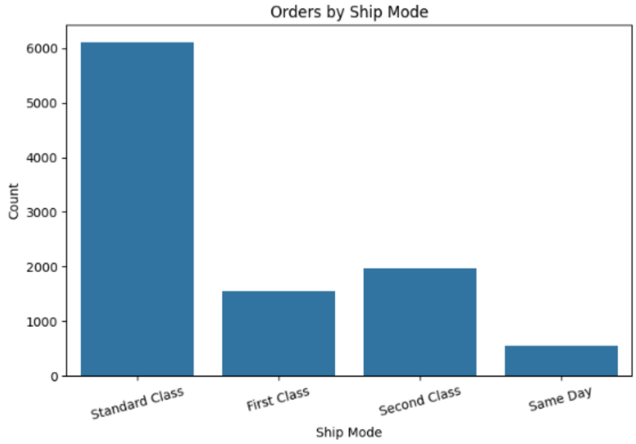


- **Scatter Plot –** Analyze relationship between sales and profit

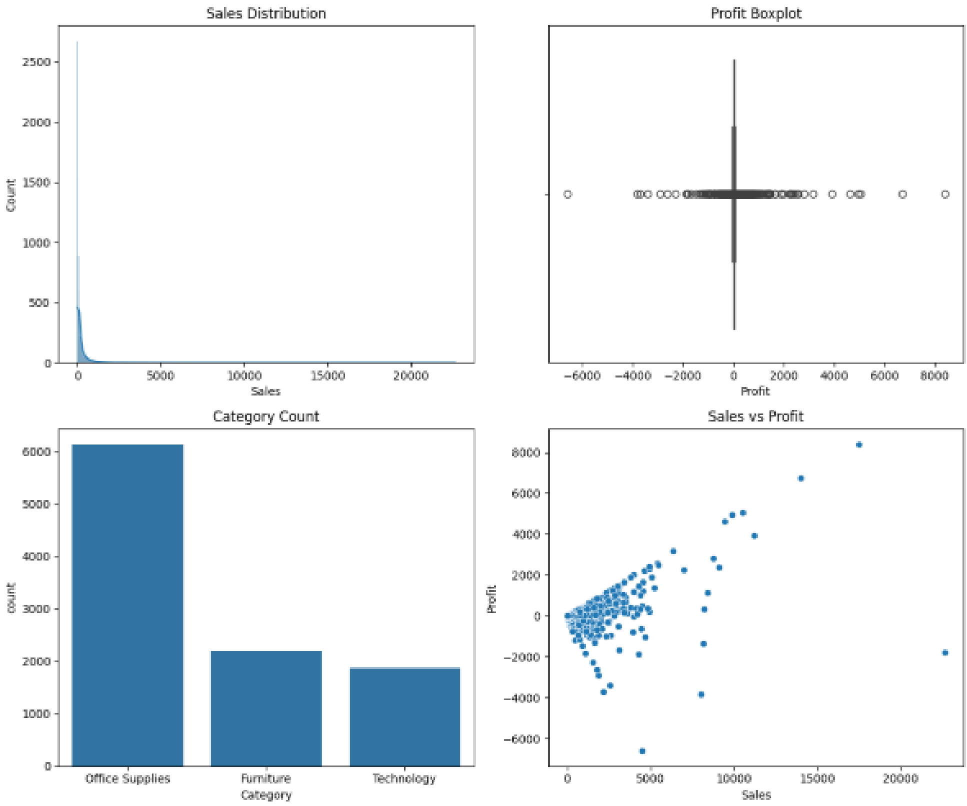
- **Heatmap –** Display correlation between numerical variables



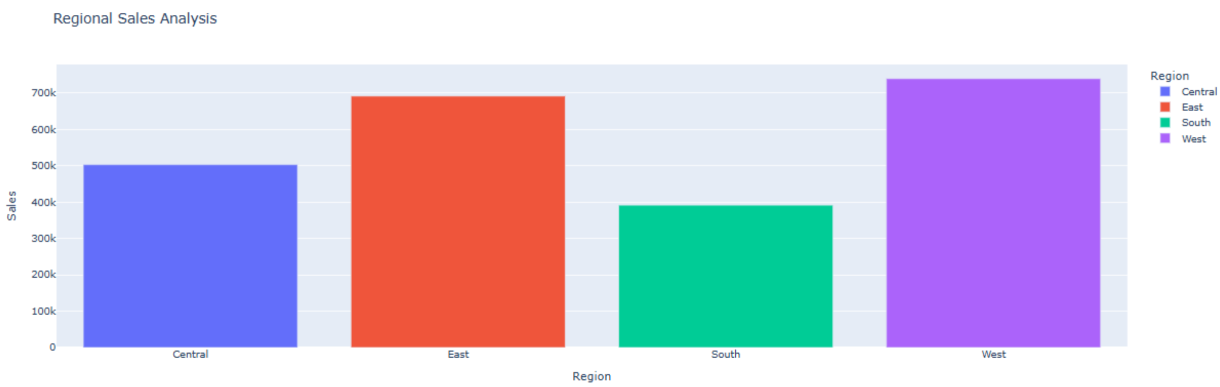
- **Count Plot –** Analyze frequency of shipping modes



- **Subplots –** Display multiple charts in a single layout



- **Plotly Interactive Charts –** Interactive regional and category analysis



7. Files Included

- superstore_data.xls – Raw dataset
- superstore_sales_main_project.ipynb – Pandas analysis and visualizations
- Superstore-Project-Documentation – Project description and usage instructions

8. How to Use

- Open superstore_sales_main_project.ipynb using Google Colab.
- Run the notebook cells step by step to view data cleaning, analysis, and visualizations.

9. Conclusion

This project demonstrates how Python libraries such as Pandas, NumPy, Matplotlib, Seaborn, and Plotly can be effectively used for real world sales data analysis and visualization. The analysis helped uncover important patterns related to sales performance, profitability, customer segments, regional trends, and discount impact. The insights gained from this project can help businesses better understand customer behavior, optimize sales strategies, improve profitability, and make informed data-driven decisions.